Lecture 7: Competition & Long-Run Production (Chapter 4)

Applied Microeconomics — Master Notes · by Jakob V. Stangel

Required reading: Friberg, **Ch. 4 — Supply by price-taking firms** (the **long-run** half: output rules, long-run shutdown & entry, the long-run market supply curve). *Lecturer: L. Butera — deck "**Competition and long-run production.**" (Lecture 6 built the firm's costs and the short-run supply rule; this lecture pushes the same firm into the **long run**, where firms can **enter and exit** — and derives the **market** supply curve.)*

Résumé. Lecture 6 left the firm making a **short-run** profit at price p by setting **MR = MC**. This lecture asks the natural follow-up: *are those profits sustainable?* In **perfect competition** the answer is **no** — because **entry is free**, any economic profit attracts new firms, output rises, and the **price is bid down** until profit hits **zero**. That pins the **long-run price to the bottom of the average-cost curve** (min AC), tells us **how much each firm makes** and **how many firms** the market supports, and makes the **long-run market supply curve flat** at min AC. We also check *why* a competitive firm's demand is nearly horizontal, and the three cases where long-run supply **slopes** instead.

> ★ **The heart of the lecture — free entry drives price to the bottom of the AC curve.**
> • **Same output rule as the short run:** every firm still produces where **MR = MC**; for a price taker that is **p = MC**.
> • **What's new is entry/exit.** In the long run *all* costs are variable and firms can come and go. **Economic profit ⇒ entry** (more firms, price falls); **economic loss ⇒ exit** (fewer firms, price rises). The only resting point is **zero economic profit.**
> • **The long-run equilibrium condition** stacks four things on top of each other: **p = MR = MC = min AC**. Price sits at the **minimum of average cost**, so the typical firm makes **zero economic profit** and produces at **efficient scale**.
> • **Market supply** is therefore **flat at min AC** (with identical firms, free entry, constant input prices): the market delivers whatever quantity consumers want, all at the same minimum-cost price.

Red thread (two questions the whole lecture answers): *(1) Once we allow firms to enter and exit, what price and quantity does a competitive market settle at? (2) How many firms does it support — and when is the long-run supply curve flat vs sloped?*

Key concepts / "modes" to use: *market structure (the 3 determinants) · perfect competition & the price taker (→ Lecture 6) · the three equivalent **output rules** · marginal profit $\pi'(q) = 0$ · MR = MC, and p = MC for a price taker · the **long-run shutdown** rule (exit if economic loss) · free entry ⇒ **zero economic profit** · the **long-run price = min AC** (found by dAC/dq = 0) · efficient scale · the **long-run market supply curve** (horizontal sum; flat at min AC) · residual (firm-level) demand elasticity $\varepsilon_A = n\cdot\varepsilon - (n-1)\cdot\eta$ · increasing- vs decreasing-cost industries.*

Market structure & the perfectly-competitive benchmark

Core: *market structure* is just **how a market is shaped**, and it predicts **how firms will behave.** Perfect competition is the extreme where every firm is a **price taker**.

What determines market structure?

Three features (from the deck): | Determinant | Question it asks | |---|---| | **Number of firms** | many tiny firms, or a few big ones? | | **Ease of entry/exit** | can new firms join (and leave) freely? | | **Product differentiation** | identical products, or can firms make theirs distinct? |

In plain terms: the more firms, the freer the entry, and the more identical the product, the closer you are to **perfect competition** (and the less power any one firm has over price).

Perfect competition — the 5 assumptions (recap from Lecture 6)

(1) a **large number** of firms · **(2) identical (homogeneous)** products · **(3) full information** (buyers know every price) · **(4) negligible transaction costs** · **(5) free entry and exit.** → The payoff of these assumptions: each firm is a **price taker** — its own demand curve is a **horizontal line at the market price**. Charge one cent more and you sell **zero** (buyers switch to an identical rival); you can already sell all you want at p, so you never charge less. *(Assumption (5), free entry/exit, is the one doing the real work **this** lecture — it's what makes the long run different from the short run.)*

★ The three output rules (how much to produce)

Core: "produce the profit-maximising quantity" can be stated **three equivalent ways** — they always give the **same** q*.

Profit is revenue minus cost:

PROFIT

$$\pi(q) = R(q) - C(q) = p \cdot q - C(q)$$

Reads: profit **π** = total revenue **R(q)** (for a price taker, price **p** × quantity **q**) minus total cost **C(q)**. · **p** = the market price the firm takes as given.

#	Output rule	What it says
1	**Profit is maximised**	pick the q at the **top** of the profit "hill" π(q).
2	**Marginal profit = 0**	at the top of the hill the **slope** is zero: π′(q) = 0.
3	**MR = MC**	the slope of profit is MR − MC, so π′(q) = 0 means **marginal revenue = marginal cost**.

Why #2 and #3 are the same thing

Differentiate profit and set it to zero (that's rule #2):

RULE #2 ⇒ RULE #3

$$\pi'(q) = R'(q) - C'(q) = MR - MC = 0 \Rightarrow \mathbf{MR = MC}$$

Reads: the slope of the profit curve is **marginal revenue minus marginal cost**; setting it to zero gives the familiar **MR = MC.** · **R'(q) = MR** = extra revenue from one more unit · **C'(q) = MC** = extra cost of one more unit.

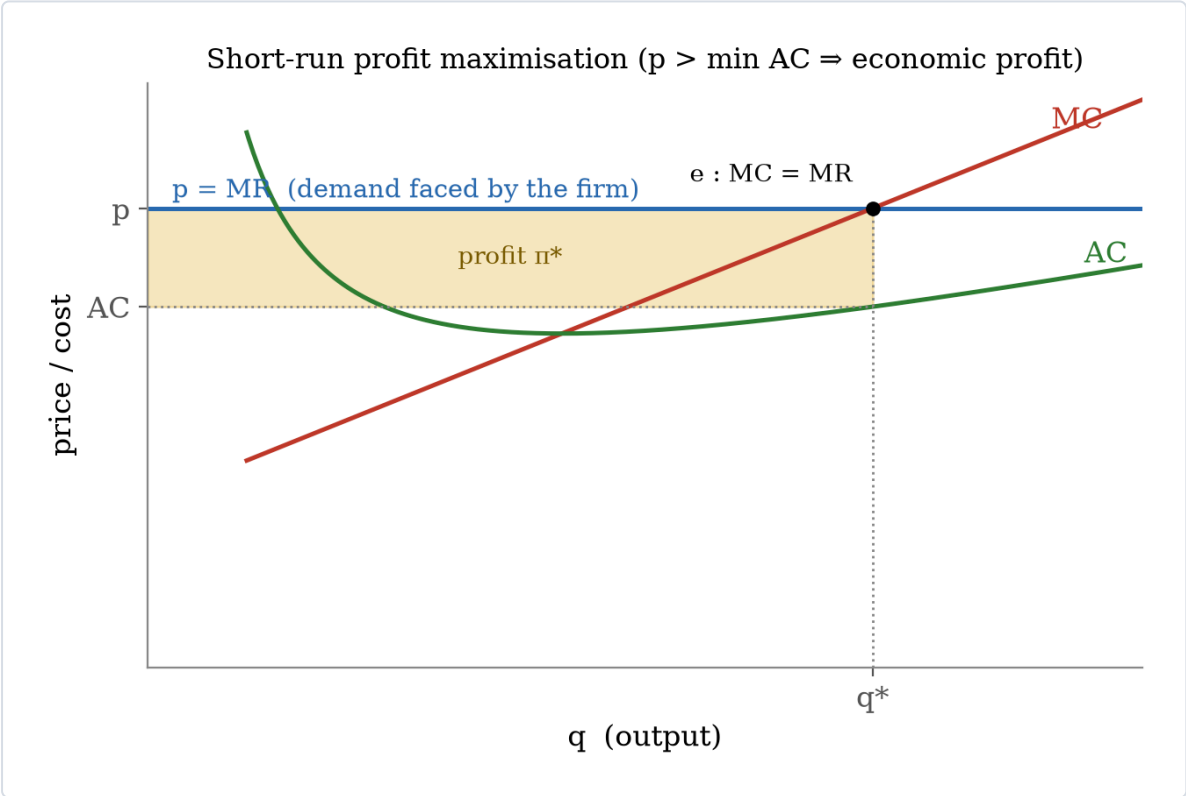
For a price taker, MR = p

Because R = p·q with *p* fixed, **MR = dR/dq = p**. So the three rules collapse to the one you already know from Lecture 6:

PRICE-TAKER'S OUTPUT RULE

$$p = MR = MC$$

Reads: a competitive firm produces the quantity where **price equals marginal cost** — that point *is* its profit-maximising output.



How to read it *(illustrative diagram)*.

- **Axes:** price/cost (y) vs output q (x).

- **p = MR** (blue) is **horizontal** — the price-taking firm's demand. **MC** (red) rises; **AC** (green) is U-shaped.
- **q*** is read off where **MC crosses p = MR** (rule #3). Rules #1 and #2 give the *same* q* at the top of the profit curve.
- The **shaded rectangle** = profit = **(p − AC)** × **q***. Here **p > AC**, so the firm makes an **economic profit** — which is exactly what the rest of the lecture says **can't last**.

Are short-run profits sustainable? (the hinge of the lecture)

Core: **no** — in the short run firms **can't** enter, so profits can persist; but the moment we allow the **long run**, free entry competes those profits away.

- In the **short run** the number of firms is **fixed** (no entry/exit), so a firm earning the shaded profit above just... keeps earning it.
- In the **long run**, that profit is a **signal**: outsiders see easy money and **enter**. More firms → more market output → the **market price falls**. Entry continues **as long as there are economic profits to be made** — i.e. until profit is **zero**.
- The symmetric case: if firms are making an **economic loss**, some **exit**, market output falls, and the price **rises** back up — again until profit is **zero**.

→ So the long-run resting point is **zero economic profit**, reached by **entry/exit moving the price**, not by any single firm changing its behaviour. *(Recall from Lecture 6: "zero economic profit" is a success — every input, including the owner's time, is earning its **opportunity cost**.)*

Long-run output & shutdown decisions

Core: the output rule is **unchanged** (MR = MC); what changes in the long run is that **all costs are variable**, which simplifies both the cost curves and the shutdown rule.

Output: same rule as the short run

The firm still picks q where **MC = MR**. Nothing new here — the rule doesn't care about the time horizon.

All costs are variable ⇒ AC = AVC

In the long run there are **no fixed costs** (you can sell the factory, not renew the lease). So **average total cost and average variable cost are the same curve**:

> LONG RUN: NO FIXED COSTS
>
> $$\text{AC} = \text{AVC} \text{ (because FC} = 0\text{, so AFC} = 0)$$

Reads: with every cost variable, total = variable cost, so **average cost (AC) equals average variable cost (AVC)** — the two SR curves merge. · *AFC = average fixed cost, which vanishes in the LR.*

Long-run shutdown: exit if you'd make a loss

Because nothing is sunk, the rule is simpler than the short run's "min AVC":

LONG-RUN SHUTDOWN RULE

$$\text{exit the market if } p < \min AC \text{ (an economic loss)}$$

Reads: in the long run a firm **leaves** the market entirely if the price is below its **minimum average cost** — because, with no sunk fixed costs, continuing would just pile up losses. · *(Contrast the **short-run** rule, min AVC: there, fixed costs are sunk, so you stay open as long as price beats **variable** cost.)*

★ The long-run price = the minimum of average cost

Core: competition drives the price **down to the lowest point of the AC curve** — so to find the long-run price, **minimise AC** (take its derivative and set it to zero).

The chain: free entry pushes the price down; it **stops** only when there's no profit left, i.e. when the price just equals the lowest possible average cost (any lower and everyone makes a loss and exits). So:

FIND THE LONG-RUN PRICE

$$\text{minimise } AC(q)\text{: } dAC/dq = 0 \Rightarrow p_{LR} = \min AC$$

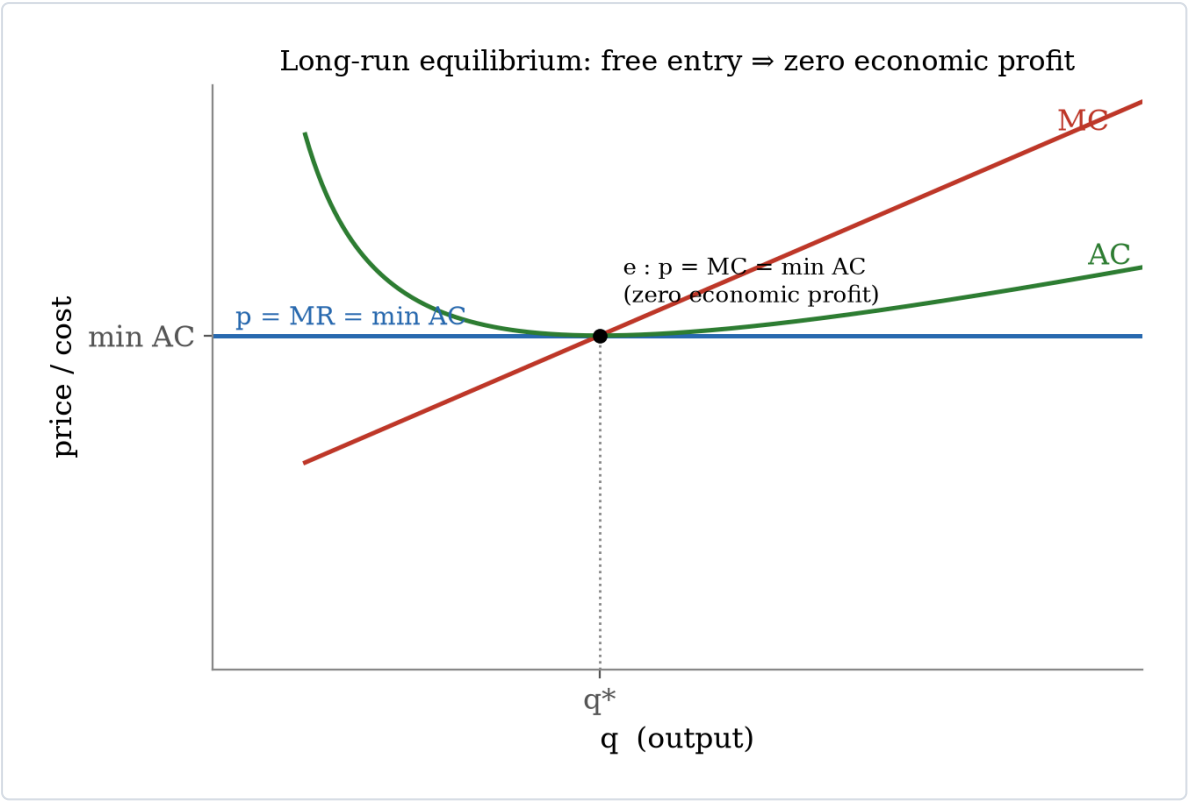
Reads: the long-run price equals the **minimum of the average-cost curve**; you find the minimising quantity by setting the **derivative of AC to zero**. · *At that minimum, **MC = AC** (marginal cuts average at its lowest point — the fact from Lecture 6).*

So the full **long-run equilibrium** stacks everything together:

LONG-RUN COMPETITIVE EQUILIBRIUM

$$p = MR = MC = \min AC \Rightarrow \pi = 0$$

Reads: in long-run equilibrium the price equals marginal revenue, marginal cost **and** the minimum of average cost all at once — so economic **profit is zero** and each firm produces at its **efficient scale** (the output that minimises average cost).



How to read it *(illustrative diagram)*.

- Compare with the previous figure: entry has pushed **p = MR** (blue) **down** until it just **touches the bottom of AC** (green).
- At that point **MC** (red) also passes through — so **p = MC = min AC**.
- The profit rectangle has **shrunk to nothing**: **zero economic profit.** The firm produces at **efficient scale** (the q that minimises AC).

★ Worked example — entry, the long-run price, and how many firms

Core: the exam-style question: given one firm's costs and the market demand, find **each firm's output**, the **long-run price**, and the **number of firms**. *(This is the deck's own example, extending Lecture 6's short-run version.)*

Given: each firm has **TC = 2q² + 50q + 50**; the short-run price was **p = 100**; **market demand is p = 210 − Q** (Q = total market quantity).

Step 1 — build MC and AC.

MARGINAL & AVERAGE COST

$$\mathrm{MC} = \mathrm{dTC/dq} = 4\mathrm{q} + 50 \quad \mathrm{AC} = \mathrm{TC/q} = 2\mathrm{q} + 50 + 50/\mathrm{q}$$

Reads: **MC** is the derivative of total cost; **AC** is total cost per unit. · the **50/q** term is the fixed cost (50) spread over output — it's what makes AC U-shaped.

Step 2 — find the long-run price = min AC (set dAC/dq = 0):

MINIMISE AC

$$\mathrm{dAC/dq} = 2 - 50/q^2 = 0 \Rightarrow q^2 = 25 \Rightarrow \mathbf{q^* = 5}$$

Reads: differentiate AC, set to zero, solve for q. Each firm's **efficient-scale output is q* = 5.** · (Second check: AC'' = $100/q^3 > 0$, so it's a minimum.)

Step 3 — the long-run price (plug $q^* = 5$ into AC):

LONG-RUN PRICE

$$p_{LR} = \min \mathrm{AC} = 2(5) + 50 + 50/5 = \mathbf{70} \ (\text{check: } \mathrm{MC} = 4(5)+50 = 70 = \mathrm{AC} \ \checkmark)$$

Reads: the long-run price is **70** — and MC equals AC there, confirming it's the minimum. · *(The SR price of 100 > 70 meant SR economic profit → that's exactly what drew firms in and dragged the price down to 70.)*

Step 4 — market quantity (from demand at p = 70):

TOTAL MARKET OUTPUT

$$p = 210 - Q \Rightarrow Q = 210 - 70 = \mathbf{140}$$

Reads: at the long-run price of 70, consumers buy **Q = 140** in total across the whole market.

Step 5 — how many firms? (market quantity ÷ each firm's output):

NUMBER OF FIRMS

$$n = Q \ / \ q^* = 140 \ / \ 5 = \mathbf{28 \text{ firms}}$$

Reads: with each firm producing 5 and the market absorbing 140, the market supports **n = 28** identical firms in the long run. · **This is the number that free entry delivers** — exactly enough firms to meet demand at zero profit.

★ The long-run market supply curve

Core: the **market** supply curve is the **horizontal sum** of all the firms' supply curves — but in the long run the **number of firms isn't fixed**, so with identical firms it comes out **flat at min AC**.

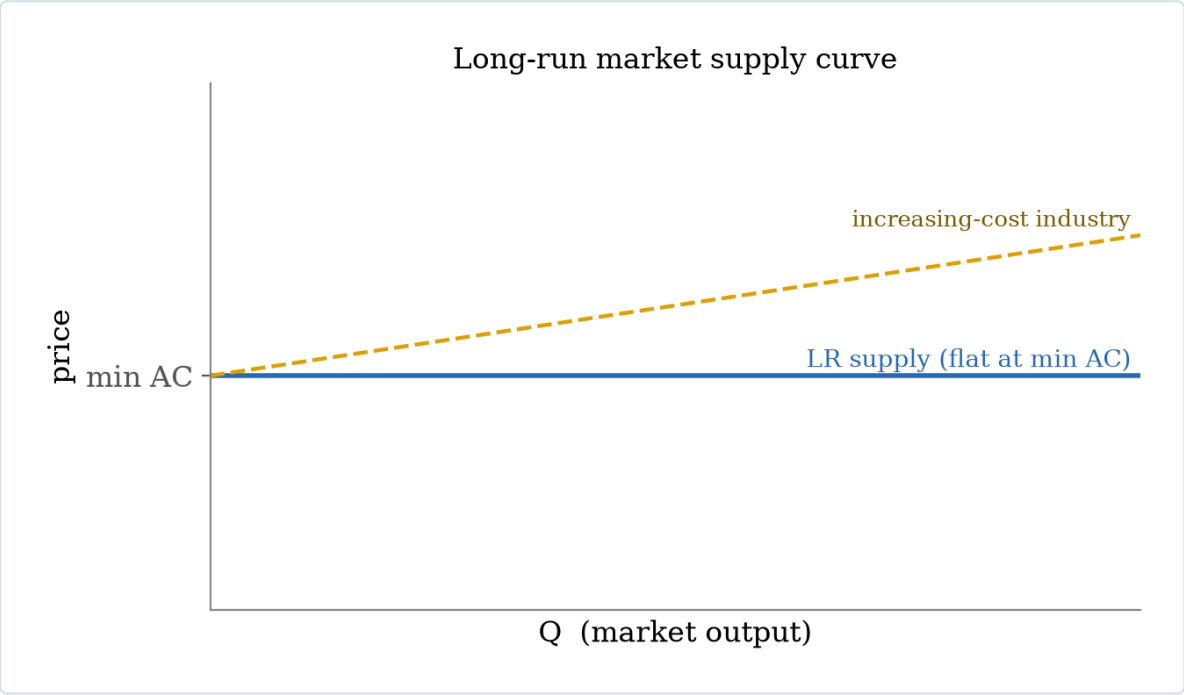
- As in the short run, market supply = **add up quantities across firms at each price** (horizontal sum).
- The twist: firms **enter** whenever a long-run profit is available and **exit** to avoid a long-run loss. So the market can supply **any** quantity at the price **min AC** — just add or subtract firms.

- With **identical firms + free entry + constant input prices**, the result is a **horizontal (flat) long-run supply curve at min LR average cost**: as new firms enter, profits fall to zero, and the price can't stay above (entry) or below (exit) **min AC**.

LONG-RUN MARKET EQUILIBRIUM (IDENTICAL FIRMS)

$$P = MR = MC = AC \text{ (at min AC)} \Rightarrow \text{long-run profit} = 0$$

Reads: the whole market settles where price equals every firm's marginal **and** minimum average cost, so **aggregate long-run profit is zero** — the market delivers consumers' desired quantity all at the one minimum-cost price.



How to read it *(illustrative diagram)*.

- **Axes:** price (y) vs **market** quantity Q (x).
- The solid **blue line is flat at min AC**: whatever quantity demand calls for, the market supplies it at that one price — because **entry/exit** adjusts the number of firms rather than the price.
- The dashed amber line previews the next section: if input prices rise with industry output (an **increasing-cost industry**), long-run supply **slopes up** instead.

Is the firm's demand *really* flat?

Core: an individual competitive firm faces a demand curve **far more elastic** (flatter) than the **market's** — flat enough to treat as horizontal — and we can show *how much* flatter.

The firm's own (**residual**) demand elasticity depends on the market elasticity, the number of firms, and how elastically the **other** firms supply:

FIRM-LEVEL (RESIDUAL) DEMAND ELASTICITY

$$\varepsilon_A = n \cdot \varepsilon - (n-1) \cdot \eta_{\text{other firms}}$$

Reads: the elasticity of demand faced by **firm A** = the **number of firms** n times the **market demand elasticity** ε, minus $(n-1)$ times the **supply elasticity of the other firms** η. · **The more firms there are, and the more elastically they supply, the more elastic (flatter) firm A's demand.**

Worked numbers (soybean, from the deck): market demand elasticity **$\varepsilon = -1.1$**, other-firm supply elasticity **$\eta = 3.1$**, **$n = 78$** firms:

SOYBEAN EXAMPLE

$$\varepsilon_A = 78 \cdot (-1.1) - 77 \cdot (3.1) = -85.8 - 238.7 \approx \mathbf{-324.5}$$

Reads: although the **whole soybean market** is only **moderately** elastic (-1.1), a **single farm** faces a **super-elastic** demand (≈ -324.5) — in effect a **flat** line. · *This is what justifies the "price taker / horizontal demand" assumption: it's an approximation, but a very good one when firms are many.*

When the long-run supply curve is *not* flat

Core: the flat LR supply curve relies on identical firms + free entry + constant input prices. **Break any of those** and the curve **slopes**.

Scenario	What breaks	LR supply shape
Entry is limited	government caps the number of firms, a **scarce resource** is needed, or **entry is costly**	**upward-sloping**
Firms differ	low-min-AC firms enter at lower prices than high-cost ones	**upward-sloping** (cheap firms first, then dearer)
Input prices vary with output — increasing-cost industry	expanding the industry **bids up** input prices	**upward-sloping**
Input prices vary with output — decreasing-cost industry	expanding the industry **lowers** input prices (e.g. supplier economies of scale)	**downward-sloping**

In plain terms: the flat curve is the special "textbook" case; the moment some firms are cheaper than others, or growing the industry changes what inputs cost, the long-run supply curve tilts.

Formula sheet (Lecture 7)

Concept	Formula	In words
Profit	$\pi(q) = p \cdot q - C(q)$	revenue minus cost
Output rule (general)	MR = MC	produce where marginal revenue = marginal cost
Output rule (price taker)	**p = MC**	…and MR = p, so price = marginal cost
Long run: no fixed costs	AC = AVC	fixed costs vanish
Long-run shutdown	exit if **p < min AC**	leave if you'd make an economic loss
Long-run price	**p = min AC** (dAC/dq = 0)	entry drives price to the bottom of AC
LR equilibrium	**p = MR = MC = min AC**, $\pi = 0$	zero profit, efficient scale
Number of firms	$n = Q / q^*$	market quantity ÷ each firm's output
Firm demand elasticity	$\varepsilon_A = n \cdot \varepsilon - (n-1) \cdot \eta$	why one firm's demand is near-flat

★ Fun facts & memorable details (Lecture 7: Competition & Long-Run Production)

> The sticky bits from the long-run lecture.

- **Profit is a self-destruct button (in competition).** The very thing every firm wants — economic profit — is the signal that **summons the competitors who compete it away.** In perfect competition, visible profit can't survive the long run.
- **"How many firms?" is just division.** Once you've found each firm's efficient output (min AC) and the market quantity at that price, the number of firms is literally **Q ÷ q***. In the worked example: 140 ÷ 5 = **28**.
- **A single soybean farm faces demand ~300× more elastic than the market.** Market elasticity −1.1, but farm A's ≈ **−324.5**. That gap is *why* "price taker" is a fair approximation even though no demand curve is *perfectly* flat.
- **The long run erases fixed costs — and simplifies the shutdown rule.** Short run: stay open above **min AVC** (rent is sunk). Long run: nothing is sunk, so you leave below **min AC**. Same firm, different rule, because of what you can still change.
- **Competition is secretly an efficiency machine.** Driving price to **min AC** means the good is produced at the **lowest possible average cost** and sold at that cost — consumers capture all the gains, firms make zero economic profit. That's the welfare punchline behind "perfect competition."